Chapter 4: Electrostatics

The following subjects will be discussed

- Coulomb's Law for Electric Force.
- Definition of Electric Field Intensity.
- Electric Field Calculations: Point Charges.
- Electric Field Calculations: Charge Distributions.
- Gauss's Law and its Applications.
- The Electric potential.

Electric Field Due to Charge Distribution: Finite Line

The differential field is given by

$$d\vec{E}_p = \frac{1}{4\pi\varepsilon}\frac{dQ}{r^2}\,\vec{a}_r = \frac{1}{4\pi\varepsilon}\frac{\rho_L\,dl}{r^2}\,\vec{a}_r \qquad dQ = \rho_L\,dl$$

$$\vec{r} = s\,\vec{a}_\rho + (z'-z)\,\vec{a}_z = s\,\vec{a}_\rho + s\tan\alpha\,\vec{a}_z$$

$$dl = dz = -s\,(\sec\alpha)^2\,d\alpha$$

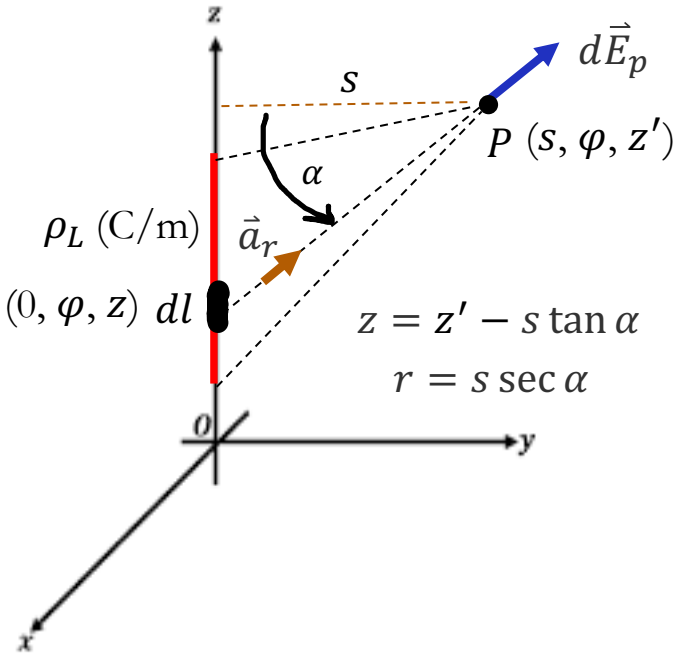
$$\vec{E}_p = \frac{1}{4\pi\varepsilon}\int_{\alpha_1}^{\alpha_2}\rho_L(-s\,(\sec\alpha)^2\,d\alpha)\frac{s\,\vec{a}_\rho + s\tan\alpha\,\vec{a}_z}{(s\sec\alpha)^3}$$

$$\vec{E}_p = \frac{-\rho_L}{4\pi\varepsilon\,s}\int_{\alpha_1}^{\alpha_2}\cos\alpha\,(\vec{a}_\rho + \tan\alpha\,\vec{a}_z)\,d\alpha$$

Assuming uniform charge density

$$\vec{E}_p = \frac{-\rho_L}{4\pi\varepsilon\,s}[(\sin\alpha_2 - \sin\alpha_1)\,\vec{a}_n - (\cos\alpha_2 - \cos\alpha_1)\,\vec{a}_L]$$

- Normal distance to axis of line
- A unit vector normal to the line
- A unit vector along the line



Special case: Infinite line.

$$\alpha_1 = 0 \qquad \alpha_2 = \pi$$

$$\vec{E}_p = \frac{\rho_L}{2\pi\varepsilon\,s}\,\vec{a}_n$$

Electric Field Due to Charge Distribution: Circular Ring

The differential field is given by

$$d\vec{E}_p = \frac{1}{4\pi\varepsilon}\frac{dQ}{r^2}\,\vec{a}_r = \frac{1}{4\pi\varepsilon}\frac{\rho_L\,dl}{r^2}\,\vec{a}_r \qquad dQ = \rho_L\,dl$$

$$r^2 = R^2 + z^2 \qquad \vec{a}_r = \vec{a}_z$$

Due to symmetry and uniformity

$$\cos\theta = \frac{z}{\sqrt{R^2+z^2}} \qquad dl = \rho\,d\varphi = R\,d\varphi$$

Cylindrical coordinates

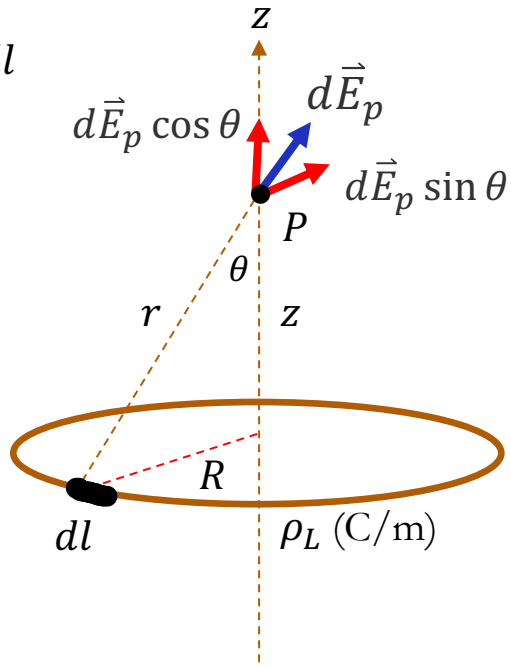
$$d\vec{E}_p = \frac{1}{4\pi\varepsilon}\frac{\rho_L R\,d\varphi}{R^2+z^2}\cos\theta\,\vec{a}_z$$

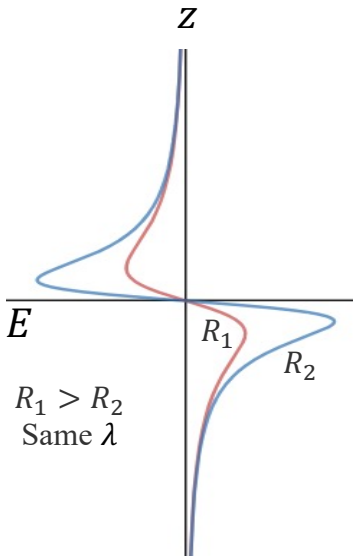
Assuming uniform charge density

$$\vec{E}_p = \frac{\rho_L}{4\pi\varepsilon}\frac{R}{R^2+z^2}\frac{z}{\sqrt{R^2+z^2}}\vec{a}_z\int_0^{2\pi} d\varphi$$

$$\vec{E}_p = \frac{\rho_L}{2\varepsilon}\frac{Rz}{(R^2+z^2)^{3/2}}\,\vec{a}_z$$

Only valid for points along the ring axis





Remarks

1) Far away from the ring, $z \gg R$

$$\vec{E}_p = k\frac{Q}{z^2}\vec{a}_z \rightarrow \text{Point charge!}$$

2) At center of the ring, $z = 0$

$$\vec{E}_p = 0 \rightarrow \text{Symmetry!}$$

Electric Field Due to Charge Distribution: Infinite Line

Example. Use the infinite line formula to find the field intensity at point $P_1(-3, 6, 8)$ and $P_2(0, 0, 4)$. The charge density is 188.5 nC/m.

$$\vec{E}_p = \frac{\rho_L}{2\pi\varepsilon\, s}\vec{a}_n$$

Point $P_1(-3, 6, 8)$

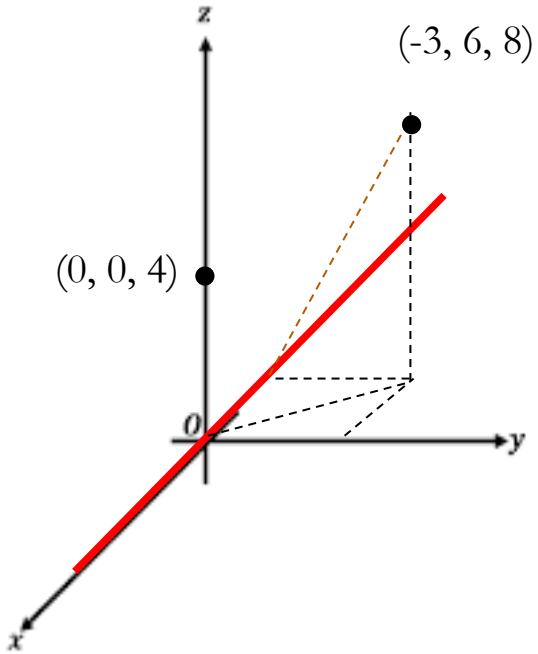
$$s = \sqrt{6^2 + 8^2} = 10 \quad \text{(m)} \qquad \vec{a}_n = \frac{6\,\vec{a}_y + 8\,\vec{a}_z}{10}$$

$$\vec{E}_{p1} = \frac{188.5}{2\pi\varepsilon\,(10)}\frac{6\,\vec{a}_y + 8\,\vec{a}_z}{10} \quad \text{(nV/m)}$$

Point $P_2(0, 0, 4)$

$$s = 4 \quad \text{(m)} \qquad \vec{a}_n = \vec{a}_z$$

$$\vec{E}_{p2} = \frac{188.5}{2\pi\varepsilon\,(4)}\vec{a}_z \quad \text{(nV/m)}$$



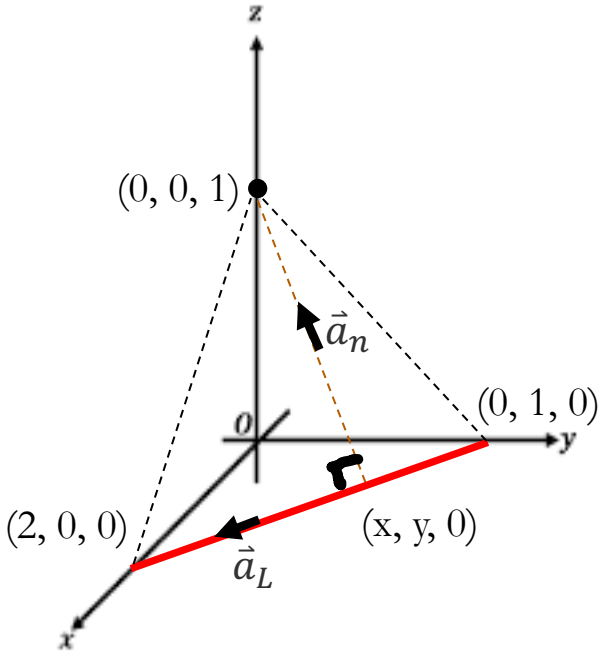
Electric Field Due to Charge Distribution: Finite Line

Example. Use the finite line formula to find the field intensity at point P. The charge density is uniform.

$$\vec{E}_p = \frac{-\rho_L}{4\pi\varepsilon\, s}\left[(\sin\alpha_2 - \sin\alpha_1)\,\vec{a}_n - (\cos\alpha_2 - \cos\alpha_1)\,\vec{a}_L\right]$$

From geometry $s = \frac{3}{\sqrt{5}}$

Find the (x, y, 0) point. (0.6, 0.7, 0)



$$\sin\alpha_1 = 1/\sqrt{10} \qquad \sin\alpha_2 = -4/5$$
$$\cos\alpha_1 = 3/\sqrt{10} \qquad \cos\alpha_2 = 3/5$$

$$\vec{a}_n = \frac{-x\,\vec{a}_x - y\,\vec{a}_y + \vec{a}_z}{\sqrt{x^2+y^2+1}} = \frac{-0.6\,\vec{a}_x - 0.7\,\vec{a}_y + \vec{a}_z}{\sqrt{0.6^2+0.7^2+1}}$$

$$\vec{a}_L = \frac{2\,\vec{a}_x - \vec{a}_y}{\sqrt{5}}$$

$$\vec{E}_p = \frac{-\sqrt{5}\rho_L}{12\pi\varepsilon}\left[\left(-\frac{4}{5} - \frac{1}{\sqrt{10}}\right)\frac{-0.6\,\vec{a}_x - 0.7\,\vec{a}_y + \vec{a}_z}{\sqrt{0.6^2+0.7^2+1}} - \left(\frac{3}{5} - \frac{3}{\sqrt{10}}\right)\frac{2\,\vec{a}_x - \vec{a}_y}{\sqrt{5}}\right]$$

Electric Field Due to Charge Distribution: Circular Disk

The differential field is given by

$$d\vec{E}_p = \frac{1}{4\pi\varepsilon}\frac{dQ}{r^2}\,\vec{a}_r = \frac{1}{4\pi\varepsilon}\frac{\rho_s\,dS}{r^2}\,\vec{a}_r$$

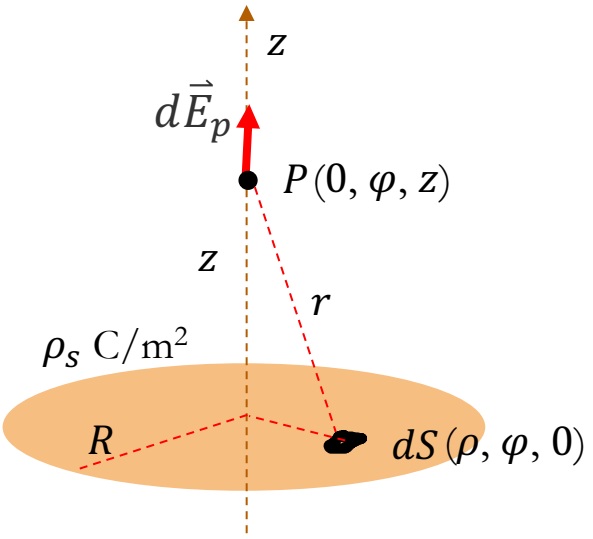
$$dQ = \rho_s\,dS \qquad dS = \rho\,d\rho\,d\varphi$$

$$\vec{r} = -\rho\,\vec{a}_\rho + z\,\vec{a}_z \qquad r = \sqrt{\rho^2 + z^2}$$

$$\vec{a}_r = \frac{-\rho\,\vec{a}_\rho + z\,\vec{a}_z}{\sqrt{\rho^2 + z^2}}$$

$$d\vec{E}_p = \frac{1}{4\pi\varepsilon}\frac{\rho_s\,dS}{\rho^2 + z^2}\frac{(-\rho\,\vec{a}_\rho + z\,\vec{a}_z)}{\sqrt{\rho^2 + z^2}}$$

Due to symmetry and uniformity



Integrate over the disk,

$$\vec{E}_p = \frac{\rho_s}{4\pi\varepsilon}\,\vec{a}_z \int_0^{2\pi}\int_0^{R} \frac{\rho}{(\rho^2 + z^2)^{3/2}}\,d\rho\,d\varphi$$

$$\vec{E}_p = \frac{\rho_s}{2\varepsilon}\left[1 - \frac{z}{\sqrt{R^2 + z^2}}\right]\vec{a}_z$$

Only valid along disk axis

$$\vec{E}_p = \frac{\rho_s}{2\varepsilon}\,z\left[\frac{1}{\sqrt{R_1^2 + z^2}} - \frac{1}{\sqrt{R_2^2 + z^2}}\right]\vec{a}_z$$

For a circular washer

Special case: Infinite plane

$$R = \infty \qquad (R \gg z)$$

$$\vec{E}_p = \frac{\rho_s}{2\varepsilon}\,\vec{a}_n$$

1) Field independent of distance.
2) Only normal component.
3) Example: parallel-plate capacitor.

Electric Field Due to Charge Distribution: Surface

Example. Find the field intensity at point P due to the given charged surface.

$$\rho_s = xy(x^2 + y^2 + 25)^{3/2} \quad (\text{nC/m}^2)$$

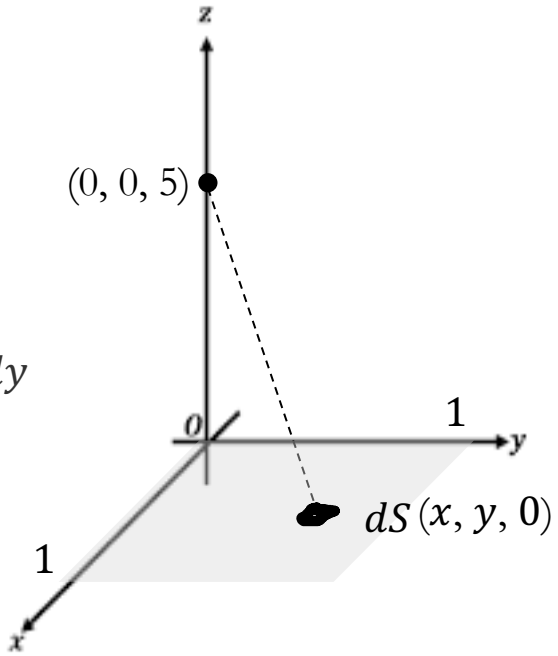
$$\vec{r} = -x\,\vec{a}_x - y\,\vec{a}_y + 5\,\vec{a}_z \qquad r = \sqrt{x^2 + y^2 + 25}$$

$$\vec{E}_p = \frac{1}{4\pi\varepsilon}\int \frac{\rho_s\, dS}{r^2}\,\vec{a}_r$$

$$\vec{E}_p = \frac{1}{4\pi\varepsilon}\int_0^1\int_0^1 \frac{xy(x^2 + y^2 + 25)^{3/2}}{(x^2 + y^2 + 25)^{3/2}}\left(-x\,\vec{a}_x - y\,\vec{a}_y + 5\,\vec{a}_z\right) dx\, dy$$

$$\vec{E}_p = \frac{1}{4\pi\varepsilon}\int_0^1\int_0^1 xy\left(-x\,\vec{a}_x - y\,\vec{a}_y + 5\,\vec{a}_z\right) dx\, dy$$

$$\vec{E}_p = -1.5\,\vec{a}_x - 1.5\,\vec{a}_y + 11.5\,\vec{a}_z \qquad (\text{V/m})$$



Gauss's Law

Gauss law relates the electric field to the charge producing it. Knowing the field, we can calculate the charge.

$$Q_{enclosed} = \psi_{net}$$

Total charge enclosed inside a 3D closed surface.

Net Flux. Number of field lines crossing a closed surface.

$$\psi_{net} = \oint \vec{D} \cdot \overrightarrow{dS}$$

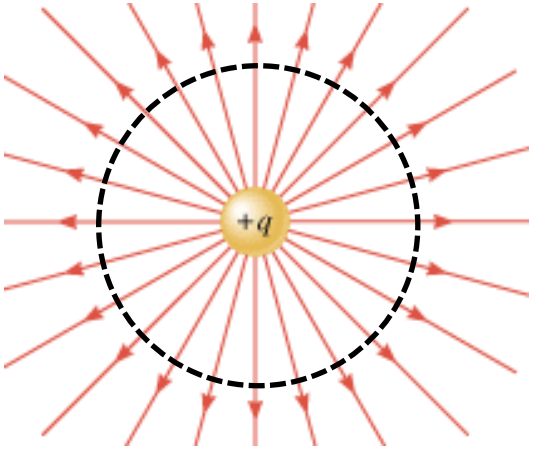
Differential area

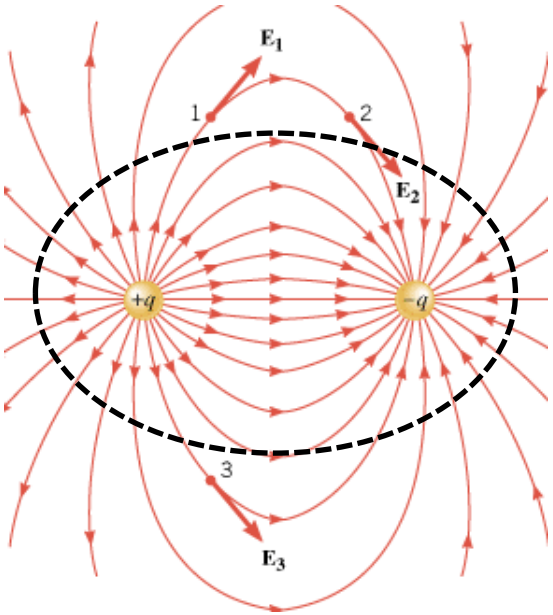
Closed surface integral

Electric flux density generated by charge

Dot product

$$\vec{D} = \varepsilon \vec{E}$$





Using divergence theorem

$$\oint_S \vec{D} \cdot \overrightarrow{dS} = \int_V \vec{\nabla} \cdot \vec{D}\ dv = Q \quad \text{and} \quad Q = \int_V \rho_v\ dv$$

The following equation is obtained

$$\vec{\nabla} \cdot \vec{D} = \rho_v$$

Flux Calculations

Example. Calculate the net electric flux out of the cube shown below given the following electric field in free space. $\vec{E} = 4\,\vec{a}_x - 3(y^2 + z)\,\vec{a}_y$ (V/m)

$$\varphi_{net} = \varepsilon_o \oint \vec{E} \cdot \overrightarrow{dS} = \varphi_{top} + \varphi_{bottom} + \varphi_{left} + \varphi_{right} + \varphi_{front} + \varphi_{back}$$

$$\varphi_{right} = \varepsilon_o \oint \vec{E} \cdot \overrightarrow{dS} = \varepsilon_o \int_0^2 \int_0^2 -3(y^2 + z)\,dx\,dz \quad (y = 2) = \varepsilon_o \int_0^2 \int_0^2 -3(4 + z)\,dx\,dz$$

$$\varphi_{right} = \varepsilon_o \int_0^2 \int_0^2 (-12)\,dx\,dz + \varepsilon_o \int_0^2 \int_0^2 (-3z)\,dx\,dz$$

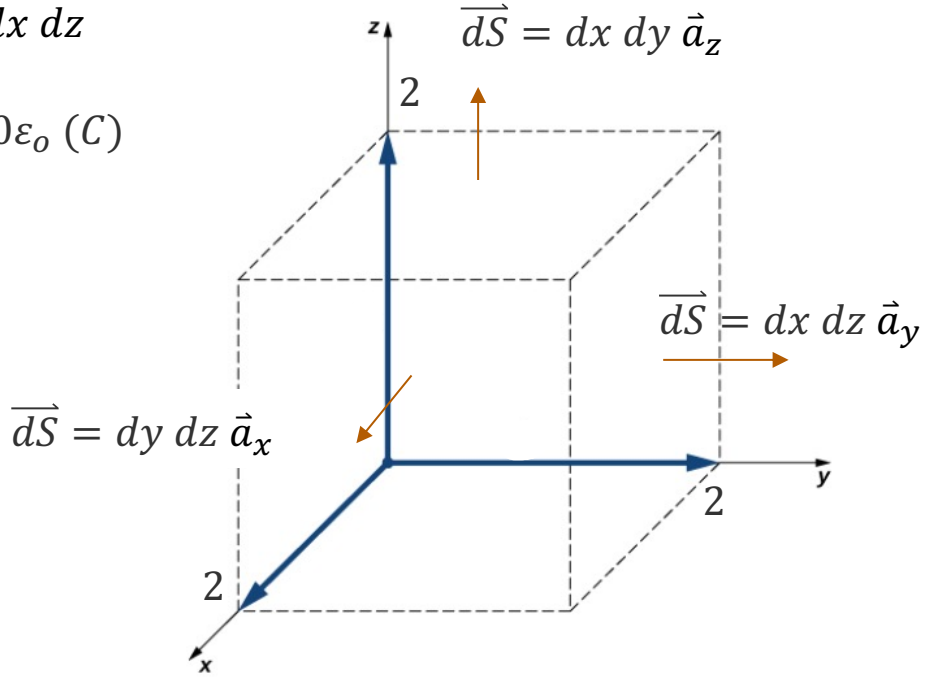
$$\varphi_{right} = \varepsilon_o(-12)(2)(2) + \varepsilon_o(-3)(2)\left(\frac{4}{2}\right) = -60\varepsilon_o\ (C)$$

$$\varphi_{left} = \varepsilon_o \int_0^2 \int_0^2 3(y^2 + z)\,dx\,dz = 12\varepsilon_o\ (C) \quad (y = 0)$$

$$\varphi_{front} = \varepsilon_o \int_0^2 \int_0^2 4\,dy\,dz = 16\varepsilon_o\ (C)$$

$$\varphi_{back} = \varepsilon_o \int_0^2 \int_0^2 -4\,dy\,dz = -16\varepsilon_o\ (C)$$

$$\varphi_{net} = -48\varepsilon_o\ (C)$$



Q & A

Send me an email if you have any questions.